

Innovative label-free and non-invasive urinary metabolite analysis integrating AI and SERS technology for cancer detection : A retrospective clinical study involving five cancer types

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Background

- Current **genomic-based liquid biopsies** often face challenges in routine clinical screening due to **high costs** and **complex procedures**.
- To address these constraints, we present AI-driven SERS signatures of urinary metabolites for **multi-cancer detection**, offering a **rapid, low-cost, and non-invasive screening approach**.

Methods

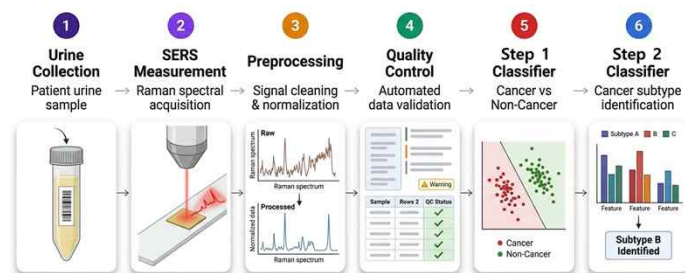


Fig 1. Analytical workflow for integrated AI-SERS technology. (1–2) SERS spectral acquisition from urine. (3–4) Data preprocessing and QC. (5–6) AI classification model for cancer detection and subtyping.

Group	n	Age	Sex (F:M)	BMI
Non-cancer	400	58 [49–65]	102:298 (26/74%)	25 [23–27]
PRC	100	72 [67–82]	0:100 (0/100%)	24 [22–26]
OVC	70	56 [47–63]	70:0 (100/0%)	24 [21–27]
LC	300	66 [60–72]	138:162 (46/54%)	24 [22–25]
PAC	70	68 [63–74]	36:34 (51/49%)	21 [19–24]
CRC	300	67 [60–76]	107:193 (36/64%)	24 [22–26]
p		<0.001	<0.001	<0.001

PRC : Prostate cancer / OVC : Ovarian cancer / LC : Lung cancer

PAC : Pancreatic cancer / CRC : Colorectal cancer

Table 1. Clinicopathological features of study participants.

Retrospectively collected urine samples from 1,240 participants across six groups (Non-cancer, n = 400; 5 types of cancer, n = 840).

Results

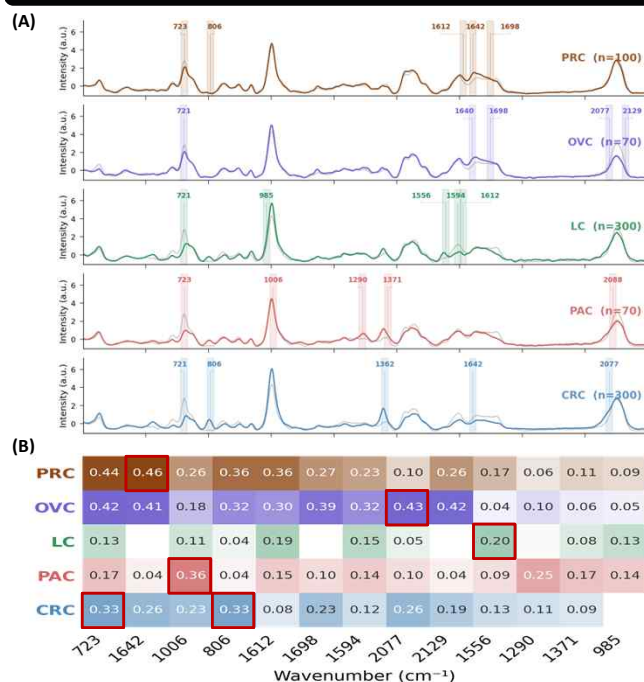


Fig 2. Cancer type-specific SERS spectral fingerprints identified by uSERS-Net (A) Discriminative peaks per cancer type reveals tumor specific metabolites signatures in urine. (B) Heatmap of discriminative peaks reveals shared yet cancer-specific intensity patterns in urinary metabolites.

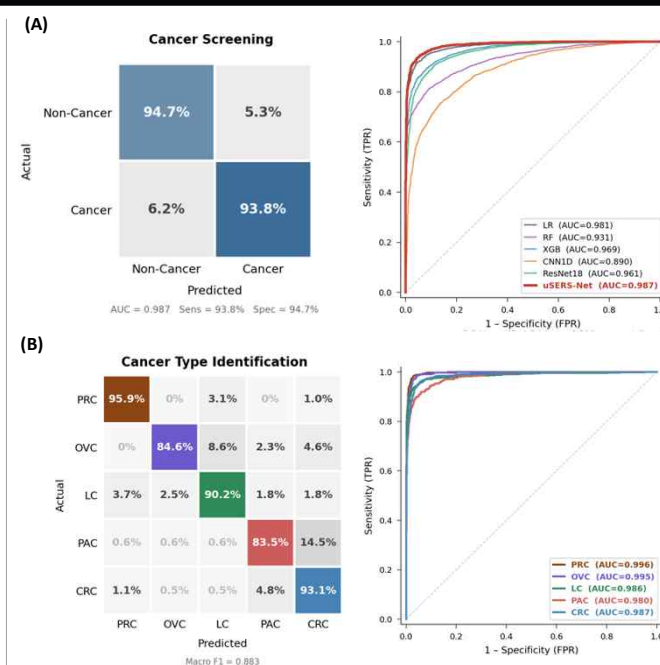


Fig 3. uSERS-Net ROC Analysis. (A) Step 1 — Cancer Screening uSERS-Net (AUC = 0.987) outperforms all baselines — a single urine test screens cancer with clinical-grade accuracy. (B) Step 2 — Cancer Type Identification All 5 cancer types classified with AUC ≥ 0.980 — screen and type in one test.

Conclusions

- As the first AI-driven urine SERS platform for multi-cancer screening, this model demonstrated exceptional performance across a large-scale cohort of **1,240 subjects from 6 institutions**, achieving an **AUC of 0.987** and a **Macro F1 of 0.883** for cancer-type identification.
- These results demonstrate that a **single, non-invasive** urine test can transcend binary cancer detection - **precisely classifying tumor origin across multiple cancer types** - establishing a robust foundation for a next-generation universal screening platform poised to transform population-level early cancer detection.

Clinical sample sources



Disclosure statement

The author declares no conflict of interest

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